

Krishi Darshan Transcript Extractor

YouTube Audio Transcription Pipeline — Project Report

annam.ai · IIT Ropar · Summer Internship 2026

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1. Project Overview

Krishi Darshan is India's longest-running agricultural television programme, broadcast by Doordarshan National since 1967. The official YouTube playlist contains 74 episodes covering diverse agricultural topics including crop techniques, animal husbandry, irrigation, biogas, and sustainable farming practices.

This project builds an automated pipeline to extract spoken content from all 74 episodes and convert it into structured, searchable text data. The resulting dataset supports downstream AI and NLP applications in annam.ai's agricultural intelligence platform.

Episodes Processed	Total Words	Avg Words / Episode
7 of 74	26,193	3,742

2. Problem Statement

Doordarshan's Krishi Darshan episodes contain decades of expert agricultural knowledge in spoken Hindi. This knowledge is inaccessible to text-based AI systems without transcription. Specific challenges encountered:

- YouTube auto-captions are not available on Doordarshan content — Whisper transcription is the only viable route.
- 74 episodes $\times$ ~25 minutes each = ~31 hours of audio, requiring a multi-session checkpoint system.
- Hindi phonetic transcription produces spelling inconsistencies requiring a post-processing normalisation pass.
- Lab VM GPU (H200) had a PyTorch/CUDA version mismatch, requiring migration to Kaggle T4.

3. Pipeline Architecture

The pipeline processes videos sequentially to manage disk space, with a checkpoint system ensuring fault tolerance across sessions.

Stage	Tool	Description
1	yt-dlp	Fetch all 74 video URLs and metadata from playlist without downloading audio
2	yt-dlp	Download audio for one video at a time as .wav file
3	Whisper-medium	Transcribe audio to Hindi text with timestamps (HuggingFace Transformers)
4	Python / CSV	Append structured row to master CSV, save checkpoint, delete audio file
5	Qwen2.5-72B (planned)	LLM post-processing pass to normalise Hindi spelling inconsistencies

3.1 Checkpoint System

A `checkpoint.txt` file stores the index of the last successfully processed video after every completed transcription. On session restart, the pipeline reads this file and resumes from the next video, appending to the existing CSV. This prevents data loss from Kaggle session timeouts (12-hour limit) or network interruptions.

3.2 Model Selection

`openai/whisper-medium` was selected over `whisper-large-v3` for Kaggle T4 compatibility. Whisper-medium processes approximately 3× faster than real-time on T4 GPU (~8 minutes per 25-minute episode) while maintaining sufficient accuracy for Hindi agricultural speech. The model runs in float16 with greedy decoding for deterministic output.

4. Results — Sample Output (7 Episodes)

ID	Title	Words	Language	Status
SxgaGXcQ...	एकीकृत कृषि प्रणाली (Integrated Farming System)	3,778	hi	✓
JLwhLr4Z...	Bhagwani ke Vikas ka Haryana me Prayas	3,342	hi	✓
ET8rwerJ...	Gaon Samridh Bharat Samridh	3,471	hi	✓
BfMmAW-5...	Chana Fasal (Chickpea Crop)	3,111	hi	✓
VzSq2eW8...	Zaed Phasal (Summer Crops)	4,021	hi	✓
tqu_7Sft...	Suksham Sichai Se Badhaye Utpadan (Micro-irrigation)	4,058	hi	✓
wlwZIC62...	Gehu ki Buaai (Wheat Sowing)	4,412	hi	✓

4.1 Transcript Quality Analysis

Whisper successfully transcribed all 7 episodes with full content coverage. The primary quality issue is phonetic Hindi spelling — agricultural terms like कृषि (krishi) appear variously as क्रिशी, किर्षी, and कुशि within the same transcript. This is a known Whisper limitation for Devanagari script and does not affect content completeness.

Episode 1 (Integrated Farming System) was manually reviewed and translated. The transcript correctly captured all key concepts including integrated farming economics, biogas plant construction and benefits, fish-duck-poultry interdependency, and government subsidy information — demonstrating the pipeline's suitability for knowledge extraction tasks.

The final Punjabi folk song at the end of Episode 1 was partially garbled, as expected — Whisper's Hindi language model performs poorly on Punjabi. This is acceptable given the primary agricultural content is in Hindi.

5. Challenges & Solutions

Challenge	Solution
HF Serverless API ignored tools= parameter for small models	Switched to JSON-mode system prompt with text output parsing
No YouTube auto-captions on Doordarshan content	Used Whisper-medium for local audio transcription
Kaggle session 12-hour limit	Checkpoint system — resumes from last completed video
Lab H200 GPU: PyTorch cu130 vs CUDA 12.7 driver mismatch	Migrated to Kaggle T4 GPU for stable inference
Session CSV/checkpoint loss between Kaggle runs	Upload both files as Kaggle Dataset between sessions
Hindi phonetic spelling inconsistency in transcripts	Planned: LLM post-processing pass (Qwen2.5-72B)

6. Next Steps

- Complete transcription of remaining 67 episodes across 2-3 Kaggle sessions
- LLM spelling normalisation pass using Qwen2.5-72B on the completed CSV
- Chunk transcripts into topic segments for vector database ingestion
- Integrate cleaned transcript dataset into Annam.ai knowledge base

7. Technology Stack

Component	Technology
Transcription Model	openai/whisper-medium (HuggingFace Transformers)
Audio Extraction	yt-dlp 2026.3.17
Audio Processing	ffmpeg, torchaudio
Compute Platform	Kaggle (T4 GPU, 16GB VRAM)
Language Runtime	Python 3.12, PyTorch 2.0+
Output Format	CSV (pandas)
Post-processing (planned)	Qwen/Qwen2.5-72B-Instruct via HuggingFace Inference API